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Theme 6: Bridging the Explainability Gap in High-Stakes Financial Decision

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About Me

- **Former Economist,** Department of Finance & Philippine Competition Commission
- Currently Marie Sklodowska-Curie Actions (MSCA) doctoral fellow in **Explainable AI and Finance**
  - Research focus: explanations for non-technical persons (e.g., public, regulators, bankers)
  - University of Naples Federico II, Italy and Swedbank Baltics
  - With CCAF
    - Co-author of 2026 Global AI Report
    - Mentor, Explainability

Workshop Agenda:

1. Welcome and Logistics
2. Basics of Explainable AI
3. Brainstorming capstone project & peer reflections
4. Wrap-up & Next Steps

1 Ice breaker

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Share one word that captures your **hope**, and one word for **concern** about AI in financial services.

Ice breaker

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2 Explainability in Financial Regulation

Explainability vs Interpretability

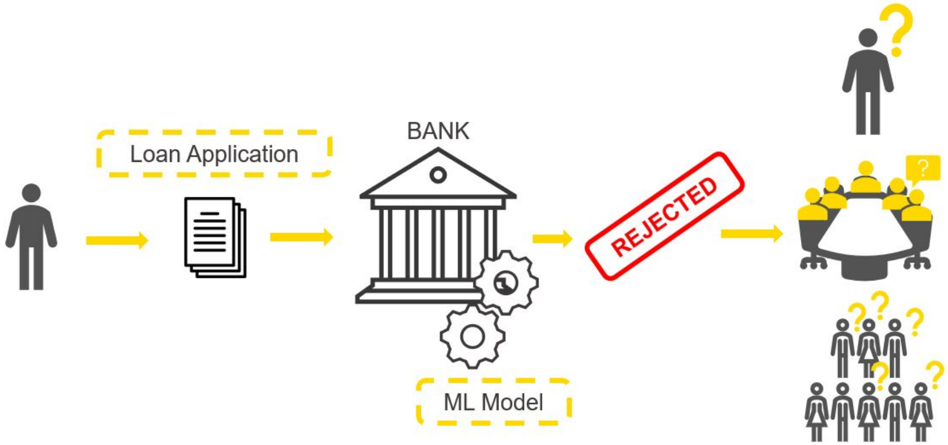
- Interpretability is about mapping an abstract concept from the models into an understandable form
- Explainability requires both interpretability and additional context

"Explainability happens with the user…and interpretability is with the (AI) tool."
(Research Participant, Europe)

- Vs. Theme 5: Transparency Disclosures and Monitoring Systems
  - Overlap for transparency purpose, but disclosures are communication obligations that inform users, customers or regulators that AI is being used, what data or purpose it serves, and what its limits are.

Why do we need explainability?

- From opaqueness of “black-box” models



- Motivation for XAI
  - Low adoption of AI
  - Method for addressing biases or supporting redress

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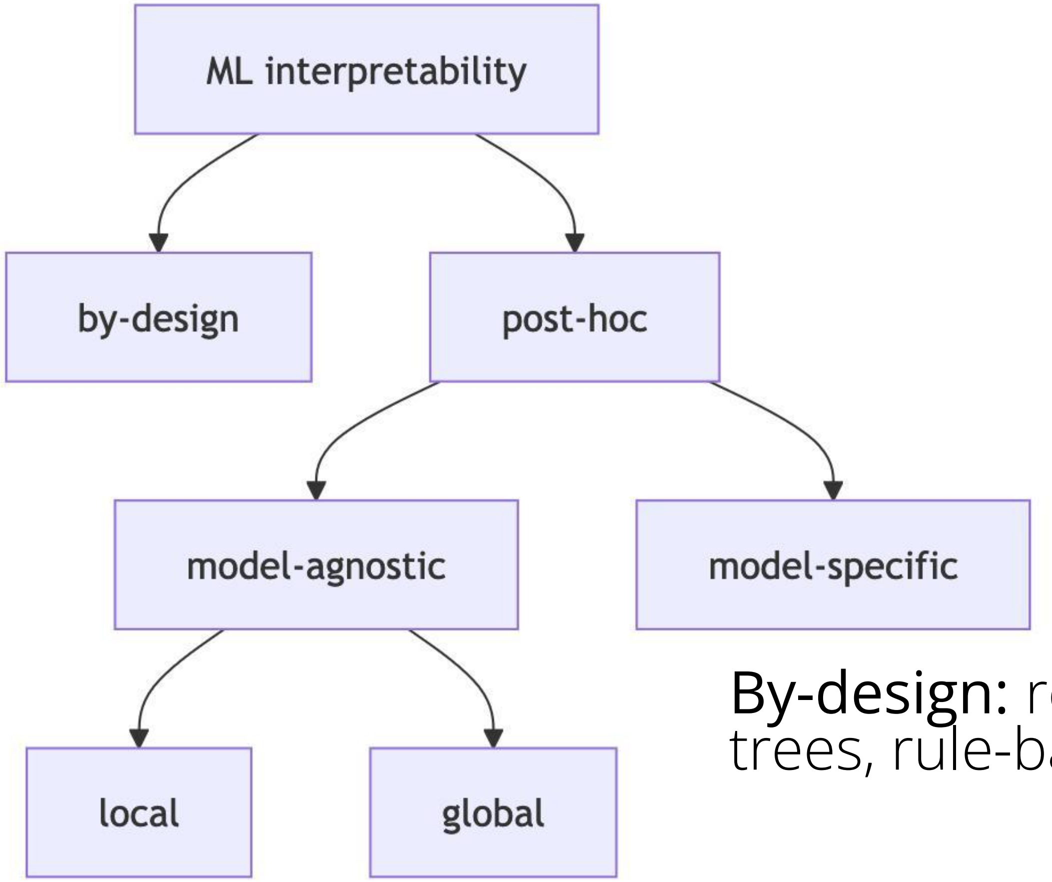
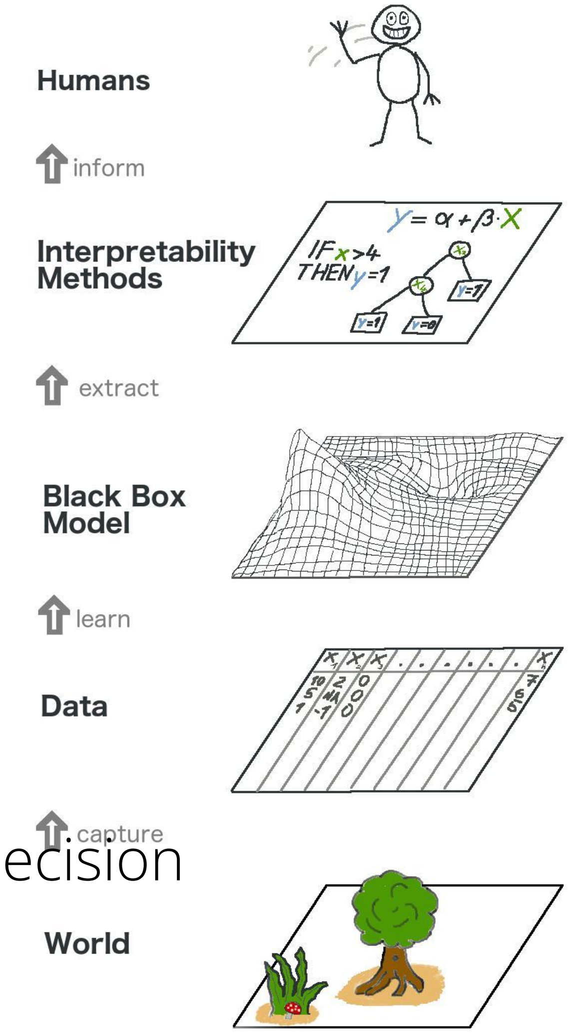
Types of XAI models



By-design: regression, decision trees, rule-based

Post-hoc: “guestimates” the AI or ML



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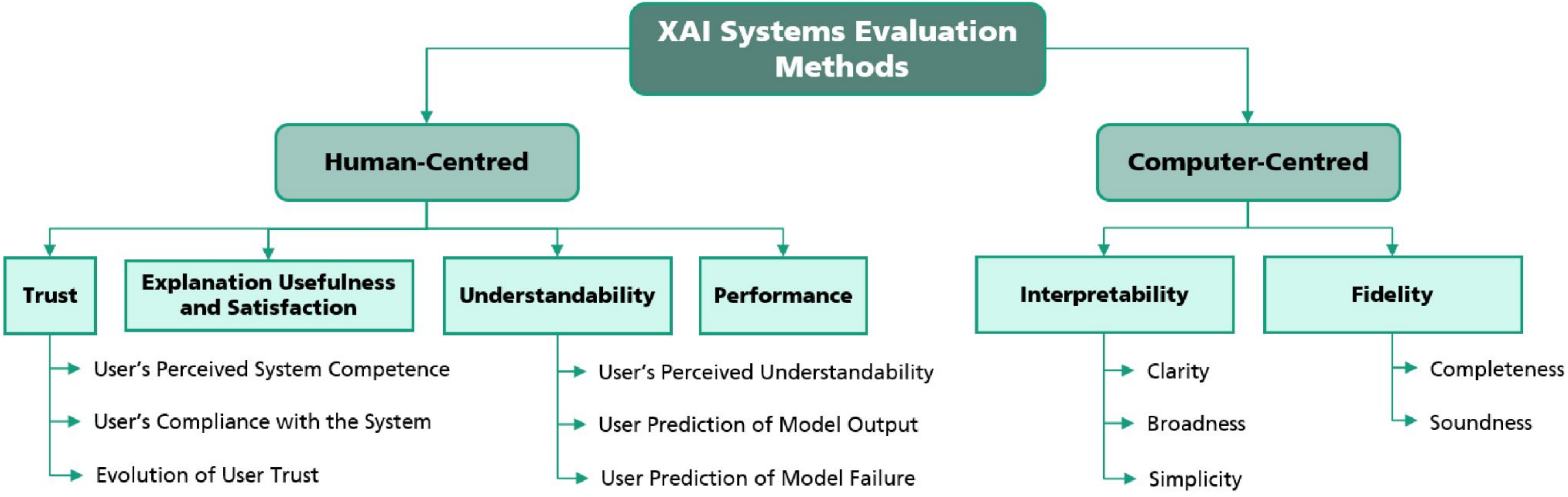
Types of XAI evaluation metrics (Lopes et al., 2022)



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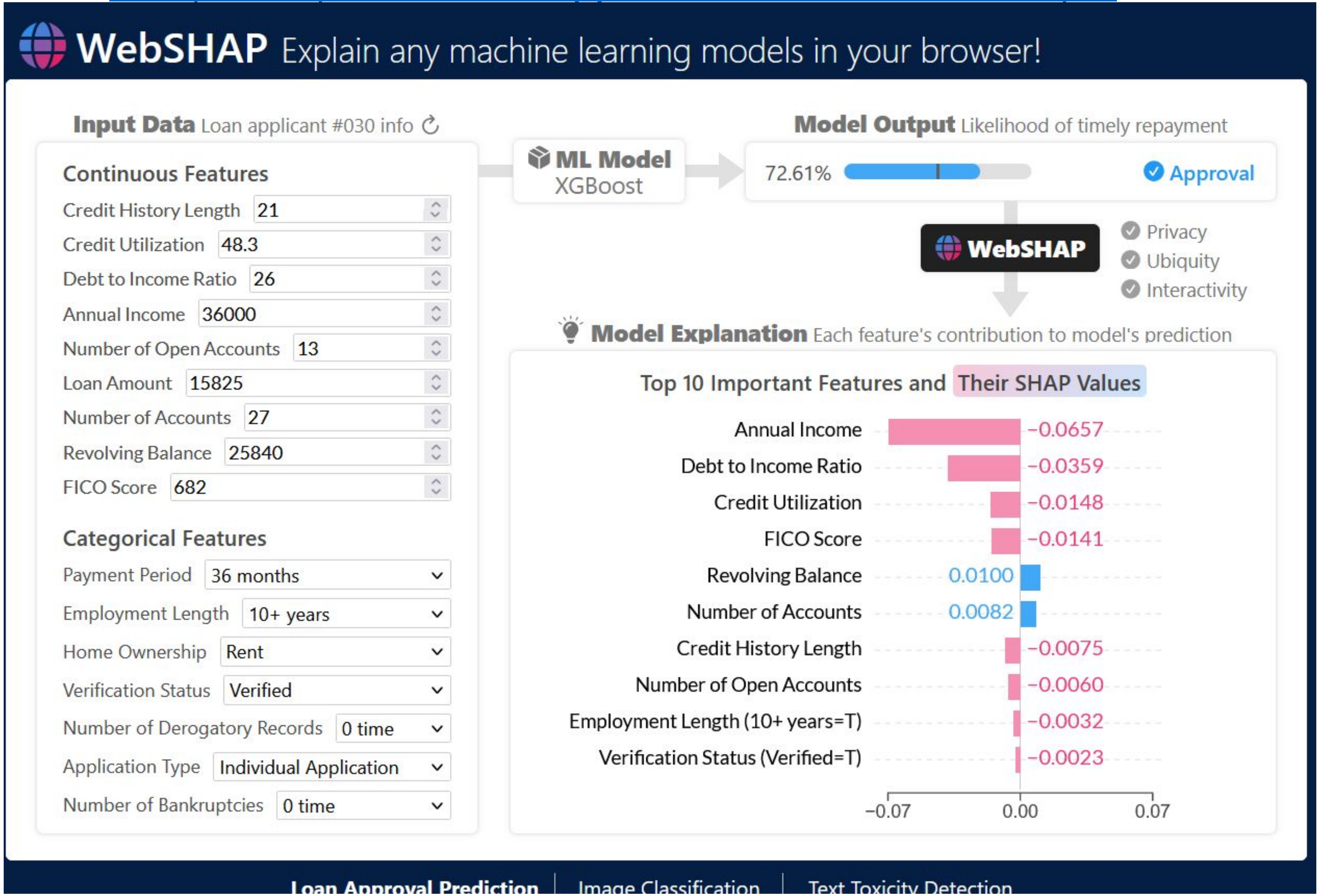
WebSHAP https://poloclub.github.io/webshap/



Open challenges: XAI is made for structured data; most documents are not

Table 8: Grad-CAM++ visualization results for cheque and signature samples

Input Images	Grad-CAM++ Output images
IDRBT dataset	
	

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Considerations as users in supervisory

- **Audience dictates the explanation format.** An explanation is only effective if the target audience can comprehend and trust it.
  - Highly complex technical visualizations, such as standard SHAP (Shapley) or bee plots, often confuse senior management and breed distrust.
  - Developers must "laymanize" outputs into simple, intuitive formats, such as basic bar graphs or normalizing Shapley values into percentage-based pie charts.
- **There are conceptual limits for XAI application in macroeconomics.** A participant reported that SHAP are poorly adapted for time-series macroeconomic data because they struggle to calculate the importance of delayed or lagged variables efficiently.

Considerations as regulators

- XAI guidance to be future-proof and at the same time, constant check-in with developments
- Prepare for redefining explainability in the age of Agentic AI

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3

Choosing a suitable capstone project

Briefly share (1-2 minutes)

1. What is the context or background? What is happening in your market, institution, or mandate that makes this important?
2. Scope: Is your idea about
    1. Internal adoption of AI in your own authority, or
    2. AI use, risks, or practices in the financial sector your supervise?
3. What type of initative are you imagining? (guidance, thematic review, checklist, sandbox theme)

Briefly react:

1. Does this idea connect with your own context or project in any way (e.g., similar topic, data needs, regulatory questions)?
2. What is one practical first step you would suggest they focus on?
3. Are there any reports, case studies you recommend them to review?

4 Wrap-up and Next Steps

Next Steps

- Finalize groupings
- Come with your draft slides In the 30-minute mentoring sessions, ideally (check appendix B for the suggested format)
- With specific concerns, can email prior the session

A Resources

Links and references

- **Interpretable Machine Learning**
  https://christophm.github.io/interpretable-ml-book/
- L. Nannini, J. M. Alonso-Moral, A. Catalá, M. Lama and S. Barro, **"Operationalizing Explainable Artificial Intelligence in the European Union Regulatory Ecosystem,"** in *IEEE Intelligent Systems*, vol. 39, no. 4, pp. 37-48, July-Aug. 2024, doi: 10.1109/MIS.2024.3383155.
  https://ieeexplore.ieee.org/abstract/document/10489989
- **Managing explanations: how regulators can address AI explainability.** Occasional Paper No. 4. BIS Financial Instability Institute.
  https://www.bis.org/fsi/fsipapers24.pdf

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B Appendix: Capstone project template

Capstone Project

🧭Structure & Template

- Use the PowerPoint template and recommended structure on the VLE

📍 Key Milestones

- **Weekly plan**: Follow the weekly capstone objectives on the VLE
- **Mentoring (Week 4/5)**: Meet with your mentor to receive early expert feedback

🚀 Submission & Sharing

- **Final submission**: Upload via the VLE submission page by **6 July 2026**
- **Optional presentation** : Present your capstone on **8 July 2026**

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AI in Financial Service for Public Authorities

[Insert Project name]

Insert group members and institution names

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Contents

Identify an ambitious, relevant and feasible topic aligned with capstone key topics that addresses a real challenge or opportunity within your authority or jurisdiction.

Section 1: Problem space articulation & context

Context to the problem:

- Briefly introduce the context, market drivers, or emerging risks making this a priority. What are the strategic regulatory or supervisory objectives of your institution in this area?
- Existing workstreams or related initiatives that this project builds on or is inspired by.

Problem space articulation:

- What is the specific AI-related challenge, risk, or operational shortcoming that needs addressing?

Crucially, specify your focus:

- Clearly state whether this addresses the *internal use and adoption of AI within your own financial authority*, or if it targets *developments, risks, or challenges within the financial industry over which you have mandate/jurisdiction*.

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Section 2: Potential approaches & implementation

Approach/Methodology:

- What actionable, high-level potential approach are you proposing to address the articulated problem?
- Examples include drafting new industry guidance, updating existing frameworks, launching a thematic review, initiating a public consultation, designing standardised reporting templates (e.g., to build an AI inventory), proposing a regulatory sandbox theme, establishing an internal cross-functional AI task force, hosting a public-private tech sprint, or scoping structured research.

Timeline for implementation:

- Briefly outline the key stakeholders involved, the resources required, and the proposed timeline or immediate first steps to launch the initiative.

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Section 3: Expected outcomes & risk management

Primary benefits:

- What concrete outcomes will be achieved? How will this initiative mitigate risks, improve oversight, inform future policy, or enhance market resilience?

Measuring success:

- How will you know if the initiative has been effective? (e.g., data gathered, guidance published, risks identified).

Initiative risks:

- What are the potential challenges or unintended consequences of implementing your proposed approach, and how will you govern or mitigate them?

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